



Special issue on technologies and applications for ubiquitous interactive media

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Modern interactive media have evolved to become reactive, context-aware, ambient, and seamlessly integrated into everyday life. These advancements have transformed how we interact with our environment, making technology an invisible yet omnipresent force in our daily routines. The challenge of developing interactive media that are both engaging and effective is inherently interdisciplinary, requiring a synthesis of human insight and technological innovation, applied research, and theoretical exploration. This editorial delves into the synergies and interdisciplinary connections necessary for the creation of new forms of interactive media.

The ambition of creating interactive media that are reactive and context-aware is a technological marvel. Such media can respond to the environment and user behavior in real-time, offering personalized experiences that are both immersive and intuitive. This responsiveness is often achieved through artificial intelligence (AI) and machine learning algorithms that analyze user data to predict needs and preferences. AI is not merely a tool but a collaborator in the design process, enabling the creation of media that can adapt and learn from interactions.

In addition to being reactive, modern interactive media are also context-aware. This means they can understand and interpret the contextual nuances of their usage environments. For instance, a smart home system can adjust lighting and temperature based on the time of day and the presence of people in the room. Similarly, mobile applications can

provide location-based services, enhancing user experience by offering relevant information and functionality. Achieving context-awareness requires an interdisciplinary approach, blending insights from fields such as human–computer interaction (HCI), cognitive psychology, and environmental design.

The ambient nature of modern interactive media ensures that technology fades into the background, becoming a seamless part of the user's environment. Ubiquitous computing, a concept that envisions computing everywhere and in everything, plays a crucial role here. Devices and sensors embedded in everyday objects and environments create an interconnected ecosystem that supports ambient interactions. This integration poses unique design and engineering challenges, necessitating collaboration across disciplines such as computer science, electrical engineering, industrial design, and even urban planning.

Furthermore, the ubiquity of interactive media implies that these technologies are embedded in various aspects of daily life, from healthcare and education to entertainment and communication. The development of such media requires not only technical proficiency but also an understanding of human behavior, social dynamics, and cultural contexts. This broadens the scope of research and design to include sociology, anthropology, and even the arts, ensuring that the solutions developed are both functional and socially relevant.

This special issue aims to explore the connections and synergies between diverse disciplines in the creation of new forms of interactive media. By bringing together contributions from domains such as artificial intelligence, virtual and augmented reality, ubiquitous computing, and human–computer interaction, we can foster new interdisciplinary connections and drive innovation. For instance, the integration of AI with virtual and augmented reality can lead to more sophisticated and responsive immersive experiences. Meanwhile, the convergence of ubiquitous computing with HCI can result in more intuitive and user-friendly interfaces. In

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total, we have received 16 contributions and 11 papers have been accepted after a thorough peer-review process.

Such interdisciplinary collaboration is not without its challenges. It requires a willingness to cross traditional academic and professional boundaries, to learn from and contribute to fields outside one's own expertise. However, the potential benefits are immense. By combining the strengths of multiple disciplines, we can create interactive media that are not only technically advanced but also deeply human-centered.

The future of interactive media lies in its ability to be reactive, context-aware, ambient, and ubiquitous. The development of such media is an inherently interdisciplinary endeavor, requiring a blend of human and technological insights, applied and theoretical research, and innovative design. By exploring and fostering connections between diverse disciplines, we can pave the way for the creation of new forms of interactive media that enhance and enrich our everyday lives. This special issue serves as a platform to bring together these varied perspectives, driving forward the exciting frontier of ubiquitous interactive media.

In summary, we are confident that this issue will be successful by leading to new applications, results, and developments. When this success materializes, it will be thanks to the many individuals who directly and indirectly

contributed to this special issue. First and foremost, we thank the authors: without their fascinating papers, this issue would not exist. Even the authors of rejected papers played a role—despite our selectivity, many of these papers contained interesting and promising ideas that, while not yet ready for publication, will likely lead to excellent future contributions.

We also owe gratitude to the referees. In a publication where papers must be comprehensible to both economists and specialists in soft computing, finding referees with expertise in both areas was challenging. Those who agreed to serve had to review more papers than usual, and their efforts are deeply appreciated. Additionally, we thank the editors of the journal for believing in our vision and for their patience with our delays. Lastly, we extend our thanks to all the readers of this issue. We appreciate those who invest the additional effort to read and understand these complex interdisciplinary papers. Thank you all, and we look forward to many more such issues in the future!

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